Calculus Review I: Limits, Completeness, Asymptotic Scales, and Local Approximation

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§1.1 Cauchy–Schwarz and Minkowski

For vectors $x, y \in \mathbb{R}^n$, the Cauchy–Schwarz inequality is

$$|\langle x, y\rangle| \le \|x\|_2 \|y\|_2.$$

If $y \neq 0$, consider the nonnegative quadratic

$$0 \le \|x - ty\|_2^2 = \|x\|_2^2 - 2t\langle x, y\rangle + t^2\|y\|_2^2.$$

Its discriminant is nonpositive, so

$$\langle x, y\rangle^2 \le \|x\|_2^2 \|y\|_2^2.$$

Equality holds exactly when x and y are linearly dependent.

Minkowski's inequality follows:

$$\begin{aligned} \|x + y\|_2^2 &= \|x\|_2^2 + 2\langle x, y\rangle + \|y\|_2^2 \\ &\le \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \\ &= (\|x\|_2 + \|y\|_2)^2. \end{aligned}$$

Hence

$$\boxed{\|x + y\|_2 \le \|x\|_2 + \|y\|_2.}$$

For nonzero vectors, equality holds exactly when x and y point in the same direction.

These inequalities control accumulated errors. If $e_1, \dots, e_m$ are perturbations, then

$$\left\| \sum_{j=1}^{m} e_j \right\| \le \sum_{j=1}^{m} \|e_j\|.$$

If an error is represented by an inner product, Cauchy–Schwarz converts it into a product of sizes. The same two patterns survive in Hilbert and normed spaces.

§1.2 Limits through Neighborhoods

For $a \in \mathbb{R}$, let

$$N_\varepsilon(a) = \{x : |x - a| < \varepsilon\}.$$

A sequence (x_n) converges to a when

$$\forall \varepsilon > 0, \quad x_n \in N_\varepsilon(a) \text{ eventually.}$$

Here "eventually" means that there exists N such that the statement holds for every $n \ge N$.

For a function f,

$$\lim_{x \to x_0} f(x) = L$$

means that for every neighborhood V of L, there is a punctured neighborhood U of x_0 such that

$$x \in U \quad \Longrightarrow \quad f(x) \in V.$$

In epsilon–delta notation,

$$0 < |x - x_0| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The neighborhood families form filters: intersections of finitely many neighborhoods are neighborhoods, and larger sets remain neighborhoods. But a *punctured* limit has a subtlety that an ordinary continuity statement does not. The tempting rule

$$f(x) \longrightarrow L, \qquad \lim_{y \to L} g(y) = M \quad \Longrightarrow \quad g(f(x)) \longrightarrow M$$

is false without an extra assumption. For instance, set $f(x) = 0$ for every x, and

$$g(y) = \begin{cases} 1, & y = 0, \\ 0, & y \neq 0. \end{cases}$$

Then $\lim_{x \to 0} f(x) = 0$ and $\lim_{y \to 0} g(y) = 0$, but $g(f(x)) = 1$ everywhere. The inner function lands exactly on the point that the outer limit *excludes*.

There are two common ways to repair the composition rule. If g is continuous at L, then $g(L) = M$, and the ordinary neighborhood definition proves $g(f(x)) \to M$. Alternatively, even if g is discontinuous at L, it suffices that $f(x) \neq L$ on some punctured neighborhood of x_0: the image then stays in the punctured neighborhood where the outer limit estimate is valid. This is why one must pay attention to which point a limit quantifier allows us to visit.

For example, the continuous outer function $g(y) = y^2$ always satisfies the familiar rule $\lim_{x \to x_0} g(f(x)) = (\lim_{x \to x_0} f(x))^2$, with no restriction on whether $f(x) = L$ along the way. In contrast, an outer function with an isolated removable discontinuity cannot be composed blindly. The filter language makes this distinction explicit: a neighborhood of L and a *deleted* neighborhood of L are different kinds of test sets.

§1.3 Uniqueness and Algebra of Limits

Suppose $x_n \to a$ and $x_n \to b$. If $a \neq b$, choose

$$\varepsilon = \frac{|a - b|}{3}.$$

For large n,

$$|a - b| \leq |a - x_n| + |x_n - b| < 2\varepsilon < |a - b|,$$

a contradiction. Thus limits are unique.

If $x_n \to a$ and $y_n \to b$, then

$$x_n + y_n \to a + b, \qquad x_n y_n \to ab.$$

For the product,

$$|x_n y_n - ab| \leq |x_n||y_n - b| + |b||x_n - a|.$$

A convergent sequence is bounded, so $|x_n| \le M$ eventually, and both terms tend to zero. If $b \neq 0$, then $y_n \neq 0$ eventually and

$$\frac{x_n}{y_n} \to \frac{a}{b}.$$

The definition of $x_n \to +\infty$ is also eventual:

$$\forall M > 0, \quad x_n > M \text{ eventually.}$$

It is not convergence to a real number, but it becomes ordinary convergence after adjoining points at infinity or applying a compactifying coordinate such as $x \mapsto \arctan x$.

§1.4 Heine's Sequential Criterion

Theorem 1.4.1 (Heine criterion)

Let f be defined on a punctured neighborhood of x_0. Then

$$\lim_{x \to x_0} f(x) = L$$

iff for every sequence $x_n \to x_0$ with $x_n \neq x_0$, one has

$$f(x_n) \to L.$$

Proof. Assume the epsilon–delta limit. Given $\varepsilon > 0$, choose $\delta > 0$ so that

$$0 < |x - x_0| < \delta \Longrightarrow |f(x) - L| < \varepsilon.$$

Since $x_n \to x_0$, the hypothesis on x_n holds eventually, so $f(x_n) \to L$.

Conversely, suppose the epsilon–delta condition fails. Then there exists $\varepsilon_0 > 0$ such that for every n one can choose x_n with

$$0 < |x_n - x_0| < \frac{1}{n}, \qquad |f(x_n) - L| \ge \varepsilon_0.$$

Then $x_n \to x_0$, but $f(x_n) \not\to L$, contradicting the sequential assumption. □

To disprove a limit, construct incompatible approaches. For

$$f(x) = \cos \frac{1}{x},$$

take

$$x_n = \frac{1}{2\pi n}, \qquad y_n = \frac{1}{(2n+1)\pi}.$$

Both tend to zero, but

$$f(x_n) = 1, \qquad f(y_n) = -1.$$

Therefore $\lim_{x \to 0} f(x)$ does not exist.

§1.5 Asymptotic Scales

Let $g(x) > 0$ near a limiting point. The notation

$$f = O(g)$$

means that there are constants $C > 0$ and a neighborhood on which

$$|f(x)| \le Cg(x).$$

The notation

$$f = o(g)$$

means

$$\frac{f(x)}{g(x)} \to 0.$$

We write

$$f = \Theta(g)$$

when both $f = O(g)$ and $g = O(f)$, and

$$f \sim g$$

when

$$\frac{f(x)}{g(x)} \to 1.$$

The implications are

$$f \sim g \Longrightarrow f = \Theta(g), \qquad f = o(g) \Longrightarrow f = O(g),$$

but the converses fail.

Equivalent infinitesimals may be substituted safely in products and quotients when the remaining factors have controlled limits. For example,

$$\sin x \sim x, \qquad 1 - \cos x \sim \frac{x^2}{2} \quad (x \to 0).$$

Hence

$$\lim_{x\to 0} \frac{1-\cos x}{x \sin x} = \lim_{x\to 0} \frac{x^2/2}{x^2} = \frac{1}{2}.$$

Substitution is not valid inside arbitrary sums where cancellation may change the leading order. Although

$$\sqrt{1+x} \sim 1,$$

one cannot replace it by 1 in

$$\sqrt{1+x} - 1.$$

The difference is of smaller scale:

$$\sqrt{1+x} - 1 = \frac{x}{\sqrt{1+x}+1} \sim \frac{x}{2}.$$

§1.6 Squeeze and monotone convergence use order structure

The squeeze theorem says that if

$$a_n \le b_n \le c_n$$

eventually and

$$a_n \to L, \qquad c_n \to L,$$

then $b_n \to L$. Indeed,

$$|b_n - L| \le \max\{|a_n - L|, |c_n - L|\}$$

when L lies between the bounding terms.

A monotone bounded sequence converges because $\mathbb{R}$ has suprema.

Theorem 1.6.1 (Monotone convergence)
If (x_n) is increasing and bounded above, then

$$x_n \to \sup_n x_n.$$

Proof. Let

$$L = \sup\{x_n : n \in \mathbb{N}\}.$$

Given $\varepsilon > 0$, the number $L - \varepsilon$ is not an upper bound. Hence some N satisfies

$$x_N > L - \varepsilon.$$

For $n \ge N$, monotonicity gives

$$L - \varepsilon < x_N \le x_n \le L.$$

Thus $|x_n - L| < \varepsilon$. □

The proof exposes the completeness input: without the least-upper-bound property, the candidate L may be missing from the number system.

§1.7 The Cauchy Criterion

A sequence is Cauchy if

$$\forall \varepsilon > 0, \quad |x_m - x_n| < \varepsilon \text{ for all sufficiently large } m, n.$$

Every convergent sequence is Cauchy by the triangle inequality.

For the converse in $\mathbb{R}$, first note that a Cauchy sequence is bounded. Choose N so that

$$|x_n - x_N| < 1 \quad (n \ge N).$$

The tail lies in a bounded interval, and the finitely many earlier terms are also bounded.

By Bolzano–Weierstrass, a bounded sequence has a convergent subsequence

$$x_{n_k} \to L.$$

Given $\varepsilon > 0$, choose K so that

$$|x_m - x_n| < \frac{\varepsilon}{2} \quad (m, n \ge K)$$

and choose k with $n_k \geq K$ and

$$|x_{n_k} - L| < \frac{\varepsilon}{2}.$$

Then for every $n \geq K$,

$$|x_n - L| \leq |x_n - x_{n_k}| + |x_{n_k} - L| < \varepsilon.$$

Thus $x_n \to L$.

The same statement fails in $\mathbb{Q}$. Let

$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n}.$$

Then (q_n) is Cauchy in the rational metric, but it has no limit in $\mathbb{Q}$. Completion adds the missing limit $\sqrt{2}$.

A Banach space is exactly a normed vector space in which every Cauchy sequence converges. Completeness is therefore not a technical appendix; it is what turns internal error control into the existence of a limit.

§1.8 Pointwise versus Uniform Continuity

A function $f : X \to Y$ between metric spaces is uniformly continuous if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad d_X(x, y) < \delta \Longrightarrow d_Y(f(x), f(y)) < \varepsilon$$

for all $x, y \in X$. The same δ must work everywhere.

The function $f(x) = x^2$ is continuous on $\mathbb{R}$ but not uniformly continuous. Set

$$x_n = n, \qquad y_n = n + \frac{1}{n}.$$

Then

$$|x_n - y_n| = \frac{1}{n} \to 0,$$

but

$$|x_n^2 - y_n^2| = 2 + \frac{1}{n^2} \not\to 0.$$

On compact domains, continuity becomes uniform.

Theorem 1.8.1 (Heine–Cantor)

A continuous function from a compact metric space to a metric space is uniformly continuous.

Proof. If uniform continuity failed, there would be $\varepsilon_0 > 0$ and pairs x_n, y_n with

$$d(x_n, y_n) < \frac{1}{n}, \qquad d(f(x_n), f(y_n)) \geq \varepsilon_0.$$

Compactness gives a convergent subsequence $x_{n_k} \to x$. Since

$$d(y_{n_k}, x) \leq d(y_{n_k}, x_{n_k}) + d(x_{n_k}, x) \to 0,$$

one also has $y_{n_k} \to x$. Continuity then forces both image subsequences to converge to $f(x)$, contradicting their separation by ε_0. □

Uniform continuity preserves Cauchy sequences and is therefore the natural hypothesis for extending a map to a completion.

§1.9 Derivatives as Linear Approximations

The derivative at a can be written as

$$f(a+h) = f(a) + f'(a)h + r(h), \qquad \frac{r(h)}{h} \to 0.$$

Equivalently,

$$f(a+h) = f(a) + f'(a)h + o(h).$$

This is stronger than the existence of a slope formula: it says the nonlinear error is negligible compared with the increment.

The differential

$$df_a(h) = f'(a)h$$

is the linear map giving the best first-order approximation. In several variables, the correct extension is the Fréchet derivative:

$$f(a+h) = f(a) + Df(a)h + o(\|h\|).$$

The derivative is therefore a local linear model, not merely a quotient of small numbers.

The chain rule follows by composing the two local models. If

$$g(a+h) = g(a) + g'(a)h + o(h)$$

and

$$f(g(a)+k) = f(g(a)) + f'(g(a))k + o(k),$$

then $k = g'(a)h + o(h) = O(h)$, so

$$f(g(a+h)) = f(g(a)) + f'(g(a))g'(a)h + o(h).$$

Thus

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

§1.10 Taylor's Theorem with Remainder

If $f \in C^{n+1}$ near a, Taylor's theorem gives

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x).$$

The Lagrange remainder has the form

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$$

for some ξ between a and x.

The integral remainder is

$$\boxed{R_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,dt.}$$

It follows by repeated use of the fundamental theorem of calculus. For example,

$$\begin{aligned} f(x) - f(a) &= \int_a^x f'(t_1)\,dt_1 \\ &= f'(a)(x-a) + \int_a^x \int_a^{t_1} f''(t_2)\,dt_2\,dt_1, \end{aligned}$$

and exchanging the order of integration produces the kernel $(x-t)$. Iterating gives the general formula.

If

$$|f^{(n+1)}(t)| \le M$$

between a and x, then

$$|R_n(x)| \le \frac{M|x-a|^{n+1}}{(n+1)!}.$$

For e^x on $[0,1]$, the degree-n Maclaurin approximation satisfies

$$\left| e^x - \sum_{k=0}^{n} \frac{x^k}{k!} \right| \le \frac{e}{(n+1)!}.$$

Taylor expansion is therefore an asymptotic statement with a certified error, not permission to replace every smooth function by an infinite series.

§1.11 Rotation matrices generate trigonometric identities

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

The matrix exponential separates into even and odd powers:

$$\begin{aligned} e^{tJ} &= I\cos t + J\sin t \\ &= \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}. \end{aligned}$$

It preserves the Euclidean quadratic form:

$$(e^{tJ})^T e^{tJ} = I.$$

The group law

$$e^{(s+t)J} = e^{sJ}e^{tJ}$$

gives the addition formulas by matrix multiplication:

$$\cos(s+t) = \cos s\cos t - \sin s\sin t,$$

$$\sin(s+t) = \sin s\cos t + \cos s\sin t.$$

Thus the identities are coordinates of the rotation group, not an isolated table to memorize.

§1.12 Lorentz boosts generate hyperbolic identities

Let

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

Then

$$\begin{aligned} e^{tK} &= I\cosh t + K\sinh t \\ &= \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}. \end{aligned}$$

Let

$$\eta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$(e^{tK})^T \eta e^{tK} = \eta.$$

Therefore the boost preserves

$$x^2 - y^2.$$

Applying it to $(1,0)^T$ yields

$$\begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix},$$

so invariance gives

$$\boxed{\cosh^2 t - \sinh^2 t = 1.}$$

Again the group law

$$e^{(s+t)K} = e^{sK} e^{tK}$$

produces

$$\cosh(s+t) = \cosh s \cosh t + \sinh s \sinh t,$$

$$\sinh(s+t) = \sinh s \cosh t + \cosh s \sinh t.$$

Circular and hyperbolic functions are parallel because both arise from one-parameter matrix groups. The sign difference comes from

$$J^2 = -I, \qquad K^2 = I,$$

and from the quadratic forms they preserve: Euclidean x^2+y^2 versus Lorentzian x^2-y^2.